MSG No. 208.46 $P4'_232'$ [Type III, cubic]

Table 1: Wyckoff site: 2a, site symmetry: 23.

No.	position	mapping
1	[0, 0, 0]	[1,2,3,4,5,6,7,8,9,10,11,12]
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[13,14,15,16,17,18,19,20,21,22,23,24]

Table 2: Wyckoff site: 4b, site symmetry: .32'

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[1,5,6,20,22,24]
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[2,10,11,16,17,21]
3	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[3,7,12,13,18,23]
4	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[4,8,9,14,15,19]

Table 3: Wyckoff site: 4c, site symmetry: .32'

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[1,5,6,20,22,24]
2	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[2,10,11,16,17,21]
3	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[3,7,12,13,18,23]
4	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[4,8,9,14,15,19]

Table 4: Wyckoff site: 6d, site symmetry: 222..

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[1,2,3,4]
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[5,8,10,12]
3	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[6,7,9,11]
4	$\left[\frac{1}{2}, 0, 0\right]$	[13,14,21,22]
5	$\left[0, 0, \frac{1}{2}\right]$	[15,16,23,24]
6	$\left[0, \frac{1}{2}, 0\right]$	[17,18,19,20]

Table 5: Wyckoff site: 6e, site symmetry: 2.2'2'

No.	position	mapping
1	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	[1,2,21,22]
2	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	[3,4,13,14]
3	$\left[\frac{1}{2}, \frac{1}{4}, 0\right]$	[5,12,18,20]
4	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	[6,9,15,24]
5	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	[7,11,16,23]
6	$\left[\frac{1}{2}, \frac{3}{4}, 0\right]$	[8,10,17,19]

Table 6: Wyckoff site: `6f`, site symmetry: `2.2'2'`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	`[1,2,21,22]`
2	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	`[3,4,13,14]`
3	$\left[0, \frac{1}{4}, \frac{1}{2}\right]$	`[5,12,18,20]`
4	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[6,9,15,24]`
5	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[7,11,16,23]`
6	$\left[0, \frac{3}{4}, \frac{1}{2}\right]$	`[8,10,17,19]`

Table 7: Wyckoff site: `8g`, site symmetry: `.3.`

No.	position	mapping
1	$[x, x, x]$	`[1,5,6]`
2	$[x, -x, -x]$	`[2,10,11]`
3	$[-x, x, -x]$	`[3,7,12]`
4	$[-x, -x, x]$	`[4,8,9]`
5	$\left[x+\frac{1}{2}, \frac{1}{2}-x, x+\frac{1}{2}\right]$	`[13,18,23]`
6	$\left[x+\frac{1}{2}, x+\frac{1}{2}, \frac{1}{2}-x\right]$	`[14,15,19]`
7	$\left[\frac{1}{2}-x, x+\frac{1}{2}, x+\frac{1}{2}\right]$	`[16,17,21]`
8	$\left[\frac{1}{2}-x, \frac{1}{2}-x, \frac{1}{2}-x\right]$	`[20,22,24]`

Table 8: Wyckoff site: `12h`, site symmetry: `2..`

No.	position	mapping
1	$[x, 0, 0]$	`[1,2]`
2	$[-x, 0, 0]$	`[3,4]`
3	$[0, x, 0]$	`[5,12]`
4	$[0, 0, x]$	`[6,9]`
5	$[0, 0, -x]$	`[7,11]`
6	$[0, -x, 0]$	`[8,10]`
7	$\left[x+\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[13,14]`
8	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}-x\right]$	`[15,24]`
9	$\left[\frac{1}{2}, \frac{1}{2}, x+\frac{1}{2}\right]$	`[16,23]`
10	$\left[\frac{1}{2}, x+\frac{1}{2}, \frac{1}{2}\right]$	`[17,19]`
11	$\left[\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}\right]$	`[18,20]`
12	$\left[\frac{1}{2}-x, \frac{1}{2}, \frac{1}{2}\right]$	`[21,22]`

Table 9: Wyckoff site: `12i`, site symmetry: `2..`

No.	position	mapping
1	$\left[x, 0, \frac{1}{2}\right]$	`[1,2]`
2	$\left[-x, 0, \frac{1}{2}\right]$	`[3,4]`
3	$\left[\frac{1}{2}, x, 0\right]$	`[5,12]`
4	$\left[0, \frac{1}{2}, x\right]$	`[6,9]`
5	$\left[0, \frac{1}{2}, -x\right]$	`[7,11]`

continued ...

Table 9

No.	position	mapping
6	$\left[\frac{1}{2}, -x, 0\right]$	[8,10]
7	$\left[x+\frac{1}{2}, 0, \frac{1}{2}\right]$	[13,14]
8	$\left[0, \frac{1}{2}, \frac{1}{2}-x\right]$	[15,24]
9	$\left[0, \frac{1}{2}, x+\frac{1}{2}\right]$	[16,23]
10	$\left[\frac{1}{2}, x+\frac{1}{2}, 0\right]$	[17,19]
11	$\left[\frac{1}{2}, \frac{1}{2}-x, 0\right]$	[18,20]
12	$\left[\frac{1}{2}-x, 0, \frac{1}{2}\right]$	[21,22]

Table 10: Wyckoff site: 12j, site symmetry: 2..

No.	position	mapping
1	$\left[x, \frac{1}{2}, 0\right]$	[1,2]
2	$\left[-x, \frac{1}{2}, 0\right]$	[3,4]
3	$\left[0, x, \frac{1}{2}\right]$	[5,12]
4	$\left[\frac{1}{2}, 0, x\right]$	[6,9]
5	$\left[\frac{1}{2}, 0, -x\right]$	[7,11]
6	$\left[0, -x, \frac{1}{2}\right]$	[8,10]
7	$\left[x+\frac{1}{2}, \frac{1}{2}, 0\right]$	[13,14]
8	$\left[\frac{1}{2}, 0, \frac{1}{2}-x\right]$	[15,24]
9	$\left[\frac{1}{2}, 0, x+\frac{1}{2}\right]$	[16,23]
10	$\left[0, x+\frac{1}{2}, \frac{1}{2}\right]$	[17,19]
11	$\left[0, \frac{1}{2}-x, \frac{1}{2}\right]$	[18,20]
12	$\left[\frac{1}{2}-x, \frac{1}{2}, 0\right]$	[21,22]

Table 11: Wyckoff site: 12k, site symmetry: ..2'

No.	position	mapping
1	$\left[\frac{1}{4}, y, \frac{1}{2}-y\right]$	[1,22]
2	$\left[\frac{1}{4}, -y, y+\frac{1}{2}\right]$	[2,21]
3	$\left[\frac{3}{4}, y, y+\frac{1}{2}\right]$	[3,13]
4	$\left[\frac{3}{4}, -y, \frac{1}{2}-y\right]$	[4,14]
5	$\left[\frac{1}{2}-y, \frac{1}{4}, y\right]$	[5,20]
6	$\left[y, \frac{1}{2}-y, \frac{1}{4}\right]$	[6,24]
7	$\left[-y, \frac{1}{2}-y, \frac{3}{4}\right]$	[7,23]
8	$\left[y+\frac{1}{2}, \frac{3}{4}, y\right]$	[8,19]
9	$\left[-y, y+\frac{1}{2}, \frac{1}{4}\right]$	[9,15]
10	$\left[\frac{1}{2}-y, \frac{3}{4}, -y\right]$	[10,17]
11	$\left[y, y+\frac{1}{2}, \frac{3}{4}\right]$	[11,16]
12	$\left[y+\frac{1}{2}, \frac{1}{4}, -y\right]$	[12,18]

Table 12: Wyckoff site: `12l`, site symmetry: `..2'`

No.	position	mapping
1	$\left[\frac{1}{4},\, y,\, y+\frac{1}{2}\right]$	`[1,21]`
2	$\left[\frac{1}{4},\, -y,\, \frac{1}{2}-y\right]$	`[2,22]`
3	$\left[\frac{3}{4},\, y,\, \frac{1}{2}-y\right]$	`[3,14]`
4	$\left[\frac{3}{4},\, -y,\, y+\frac{1}{2}\right]$	`[4,13]`
5	$\left[y+\frac{1}{2},\, \frac{1}{4},\, y\right]$	`[5,18]`
6	$\left[y,\, y+\frac{1}{2},\, \frac{1}{4}\right]$	`[6,15]`
7	$\left[-y,\, y+\frac{1}{2},\, \frac{3}{4}\right]$	`[7,16]`
8	$\left[\frac{1}{2}-y,\, \frac{3}{4},\, y\right]$	`[8,17]`
9	$\left[-y,\, \frac{1}{2}-y,\, \frac{1}{4}\right]$	`[9,24]`
10	$\left[y+\frac{1}{2},\, \frac{3}{4},\, -y\right]$	`[10,19]`
11	$\left[y,\, \frac{1}{2}-y,\, \frac{3}{4}\right]$	`[11,23]`
12	$\left[\frac{1}{2}-y,\, \frac{1}{4},\, -y\right]$	`[12,20]`

Table 13: Wyckoff site: `24m`, site symmetry: `1`

No.	position	mapping
1	$[x,\, y,\, z]$	`[1]`
2	$[x,\, -y,\, -z]$	`[2]`
3	$[-x,\, y,\, -z]$	`[3]`
4	$[-x,\, -y,\, z]$	`[4]`
5	$[z,\, x,\, y]$	`[5]`
6	$[y,\, z,\, x]$	`[6]`
7	$[-y,\, z,\, -x]$	`[7]`
8	$[-z,\, -x,\, y]$	`[8]`
9	$[-y,\, -z,\, x]$	`[9]`
10	$[z,\, -x,\, -y]$	`[10]`
11	$[y,\, -z,\, -x]$	`[11]`
12	$[-z,\, x,\, -y]$	`[12]`
13	$\left[x+\frac{1}{2},\, \frac{1}{2}-z,\, y+\frac{1}{2}\right]$	`[13]`
14	$\left[x+\frac{1}{2},\, z+\frac{1}{2},\, \frac{1}{2}-y\right]$	`[14]`
15	$\left[z+\frac{1}{2},\, y+\frac{1}{2},\, \frac{1}{2}-x\right]$	`[15]`
16	$\left[\frac{1}{2}-z,\, y+\frac{1}{2},\, x+\frac{1}{2}\right]$	`[16]`
17	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$	`[17]`
18	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$	`[18]`
19	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$	`[19]`
20	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, \frac{1}{2}-z\right]$	`[20]`
21	$\left[\frac{1}{2}-x,\, z+\frac{1}{2},\, y+\frac{1}{2}\right]$	`[21]`
22	$\left[\frac{1}{2}-x,\, \frac{1}{2}-z,\, \frac{1}{2}-y\right]$	`[22]`
23	$\left[z+\frac{1}{2},\, \frac{1}{2}-y,\, x+\frac{1}{2}\right]$	`[23]`
24	$\left[\frac{1}{2}-z,\, \frac{1}{2}-y,\, \frac{1}{2}-x\right]$	`[24]`